SCHOOL OF ENGINEERING AND NATURAL SCIENCES

EEE

Analog circuit design project

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Introduction :

This project includes the design of schematic and layout of a differential-input, single-output operational transconductance amplifier satisfying minimum specs specified.

Supply Voltage	3.3 V
Process	65 nm CMOS
Process Corner	Nominal
Temperature	27 °C
Maximum Total Current Consumption	3 mA
Min DC Gain	80 dB
Minimum Unity Gain Bandwidth	300 MHz
Minimum Phase Margin	60°
Output voltage range	0.65 - 2.65 V
Maximum settling time	(0.01%) 10 ns
Load Capacitance	3 pF
Transistor VOV	0.15 - 0.5 V

Design & methodology :

In order to meet the specs specified with the high gain, with the full output swing, and considering the maximum current consumption, based on that the symmetrical OTA was chosen because it is most likely to reach all the specs.

First some initial calculations have to be made to get an idea about the parameters and topology of the amplifier.

The first step is to determine α , β & ɣ
Firstly ɣ must be =2 for P.M. > 60° and it follows that *f*nd must be 600MHz. Then let's assume that alpha = 2 so Cc = 1.5pF and beta = 3.

The technology used has a maximum (selfgain)ftmax = 4000MHz, however when we calculate ft6>= α * ɣ(1+ β)GBW we get 4800MHz, this result is concerning making it questionable if the technology can achieve this...

In order to achieve better gain without sacrificing the output swing (cascode) for a symmetrical OTA another output stage can be added with a compensation capcitor which also eliminates the zero and helps us reach the required P.M.

The GBW then becomes: GBW = Bgm1/2 π Cc, and assuming Cc= 1.5pF & B = 3, gm1 is calculated to be 1.4mA/V, and sense we need as much gain as possible the length of the gate will be set to the minimum which is 0.5um, after that calculating the current running through the differential pair becomes clear, since gm = 2Id/Vov and from characterization of the technology we know Vov is about 0.2V then Id for each differential transistor is 140uA.
At this point we can approximate the current running through each branch and start with the schematic and initial testing there after the parameters are fine tuned to get the desired resutls.

Schematic design & device parameters :

Starting the design, by first placing the transistor and then connecting them starting with the biasing network.

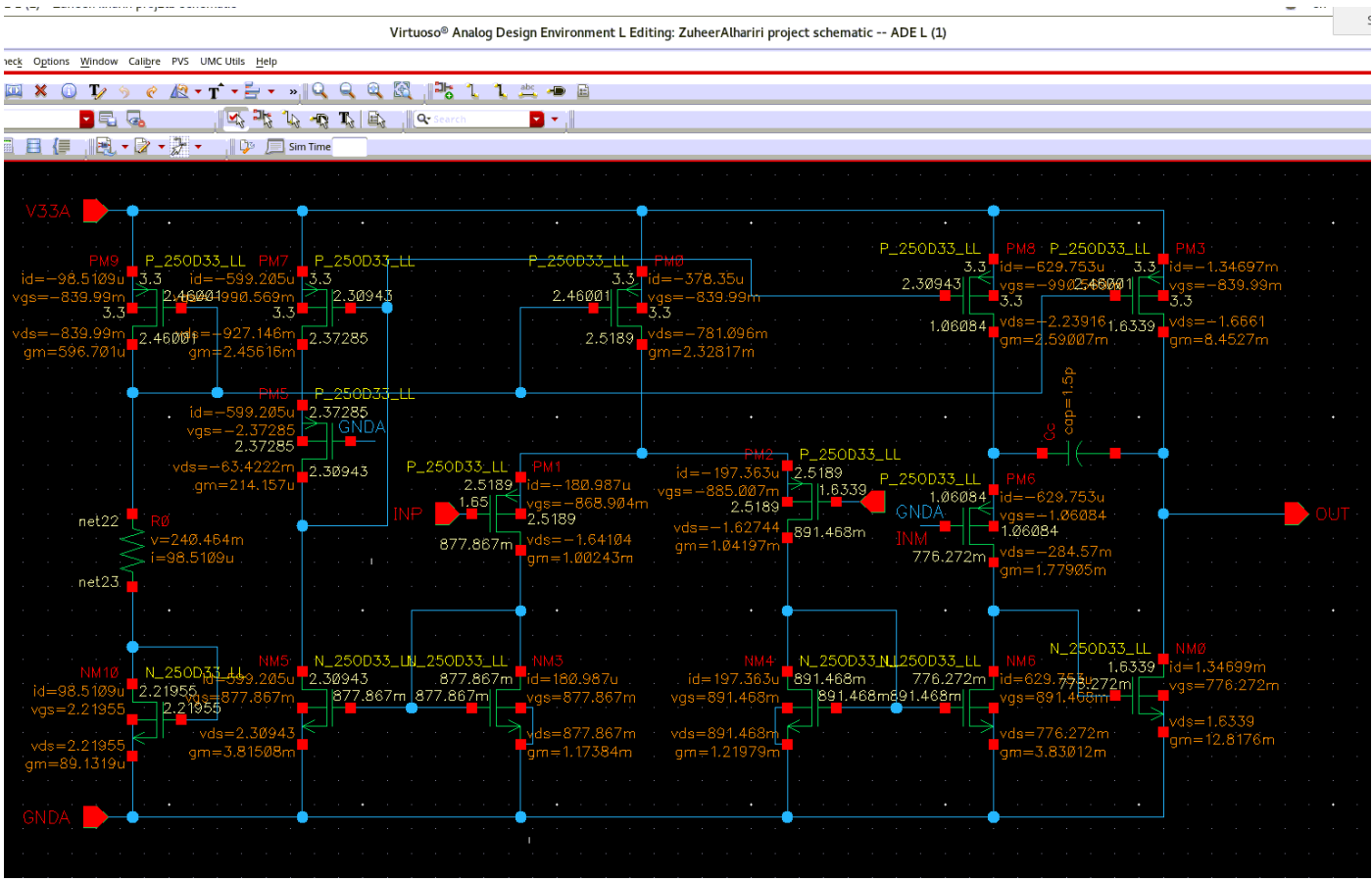

figure 1: schematic drawing in cadence

The figure above shows the circuit and transistor DC operating point after fine tuning to meet the criteria.

Device name	total width	length	fingers and multiplier
PM0	160um	0.5um	2, x1
PM1	30um	0.5um	1, x1
PM2	30um	0.5um	1, x1
PM3	420um	0.5um	5, x2
PM5	60um	0.5um	2, x1
PM6	60um	0.5um	2, x1
PM7	60um	0.5um	2, x1
PM8	60um	0.5um	2, x1
PM9	20um	0.5um	1, x1
NM0	210um	0.5um	5, x1
NM3	15um	0.5um	1, x1
NM4	15um	0.5um	1, x1
NM5	45um	0.5um	1, x1
NM6	45um	0.5um	1, x1
NM10	0.4u	0.5um	1, x1
R0	0.7u	1.9u	Resistance = 2.4k Ω
Cc	- 5 metal layers	26um, 15fingers, x10	Capacitance = 1.5fF

<u>Initial simulation & fine tuning :</u>

In the initial testing the device didn't reach the required GBW at 300MHz where it was only reaching 230MHz and the phase margin was a bit low as well, so in order to get more gain and phase parameter sweep was conducted first on the Cc and later on the length of the device, L1 length of differential pair, L2 length of mirror stage, L3 length of the output device. Changing the first two length didn't result in any better results however sweeping L3 showed that L3 = 700nm has the best gain without sacrificing the phase margin too much, however still it was not enough to reach the desired unity gain, so the next step was to try to push the first pole away and second pole away as much as possible which was done by increasing the current in the differential pair from 140uA to 200uA, and then increasing the size of pmos devices in the mirrored branches to get higher transconductance and therefore higher pole frequency.

The rest of the current was pumped into the last stage to get as much gain as possible.

Increasing the mirrored branches gm helped the phase margin recover at that point Cc parameter sweep was conducted one last time and settled on a value of 1.5pF.

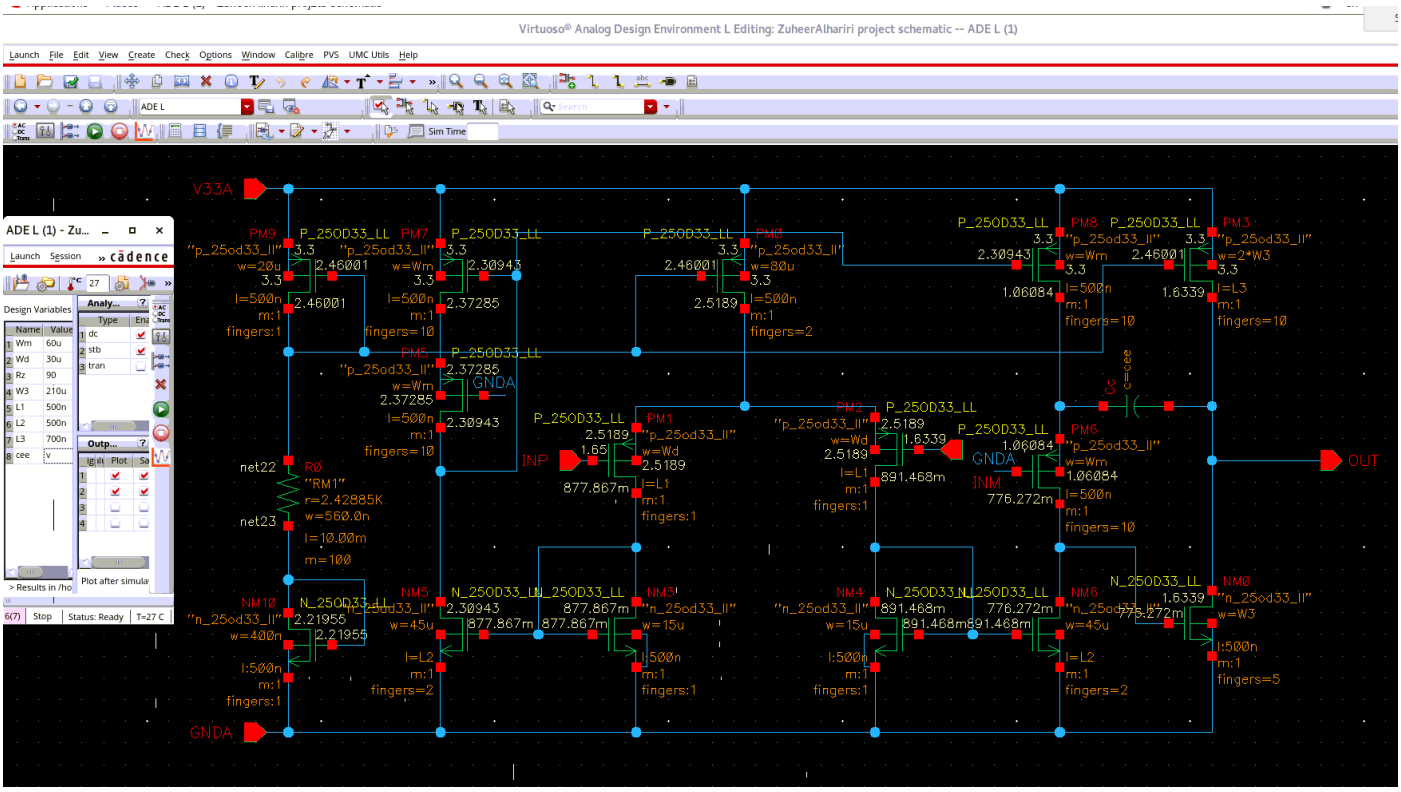

figure 2: parameters changed and initial testing & tuning

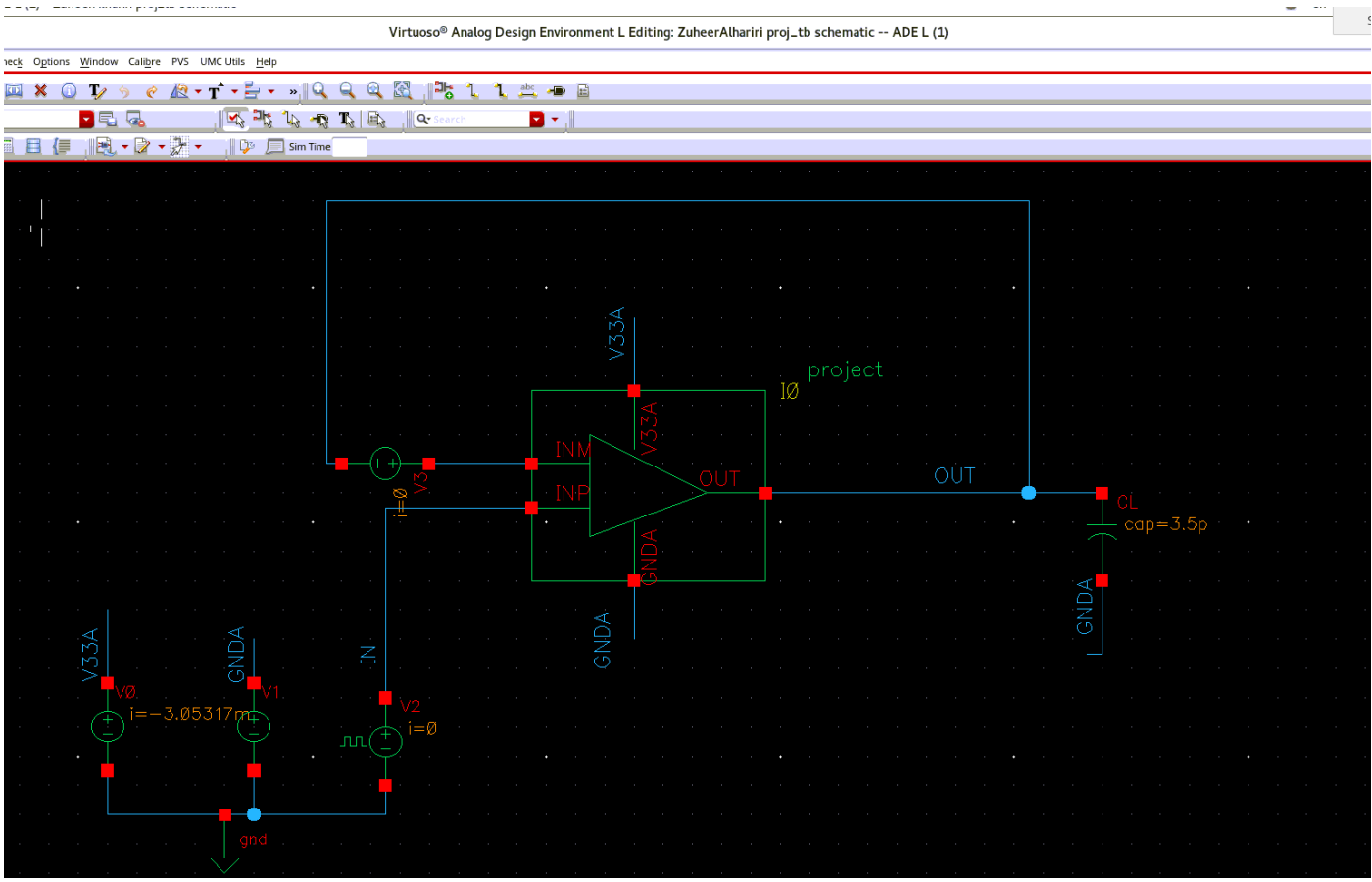

Figure 3: initial testbench-schematic

During the testing the load capacitance CL was increased to 3.5pF to similate the parasitic capacitance after extracting the PEX, it will later be turned back to 3pF for final ressult demonstration.

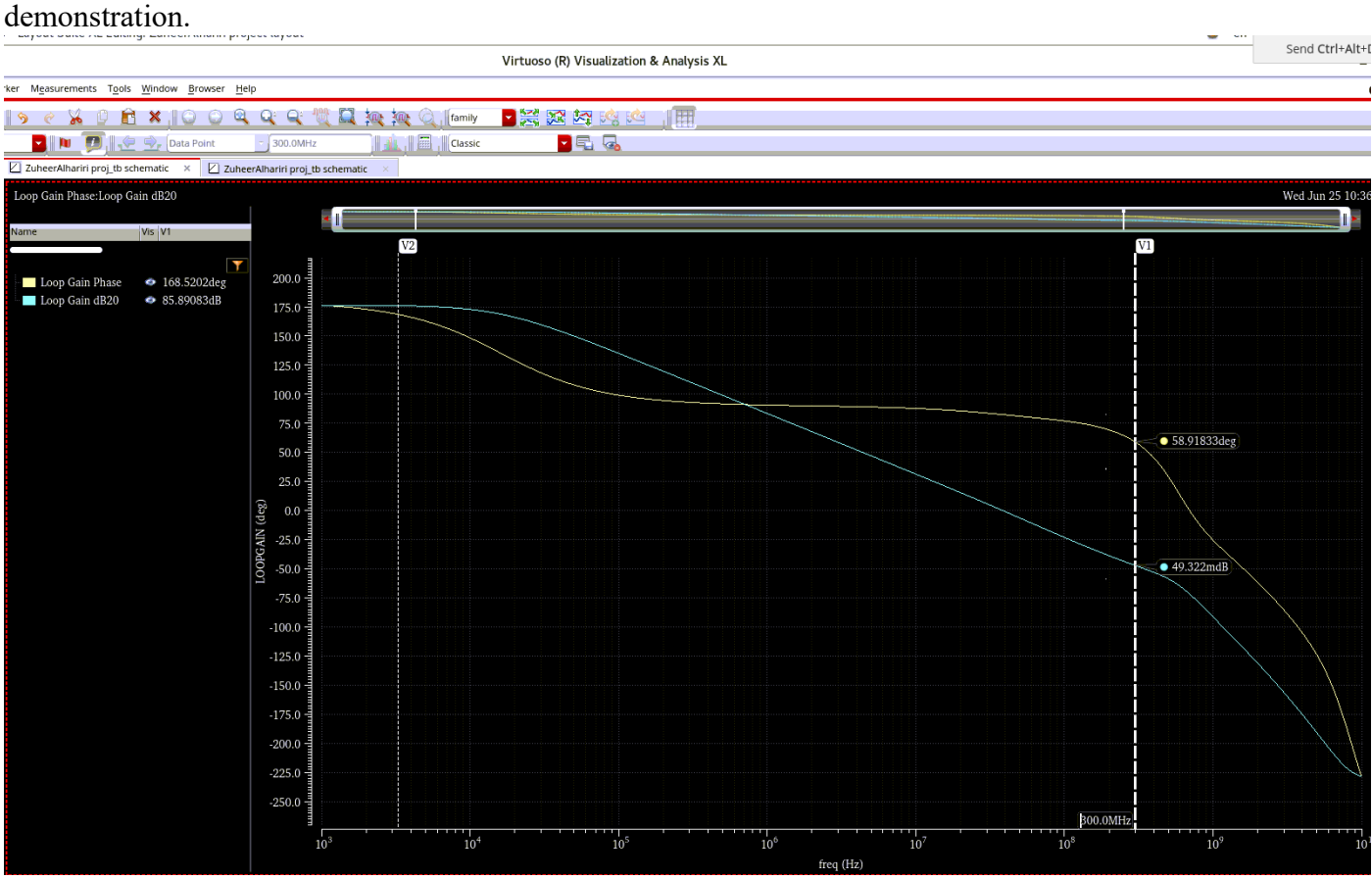

figure 4: initial testing showing STB analysis

Here the stability analysis shows at 300MHz we get 50mdB gain and 59 degree P.M.

With an open-loop gain of 85dB.

Layout design :

The next step is to start the layout design.

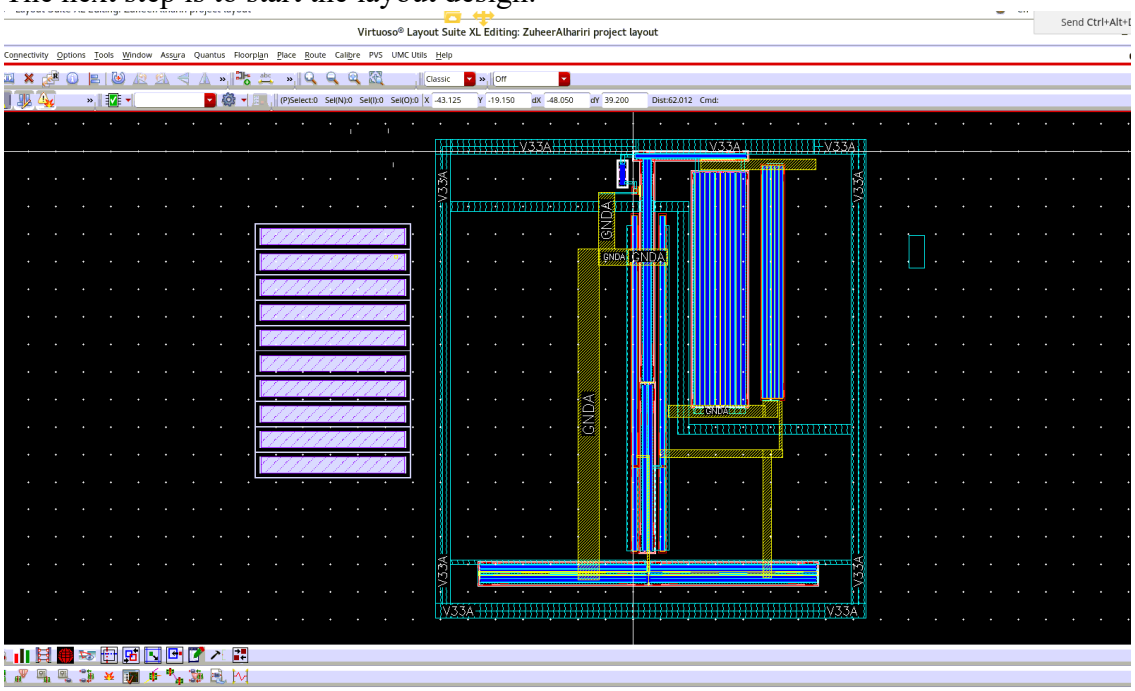

figure 5: initial layout design

Unfortunately due to none matching width/finger sizes and no pre-layout plan, the layout came out very messy and unorganized, this resulted in a lot of errors when running DRC check.

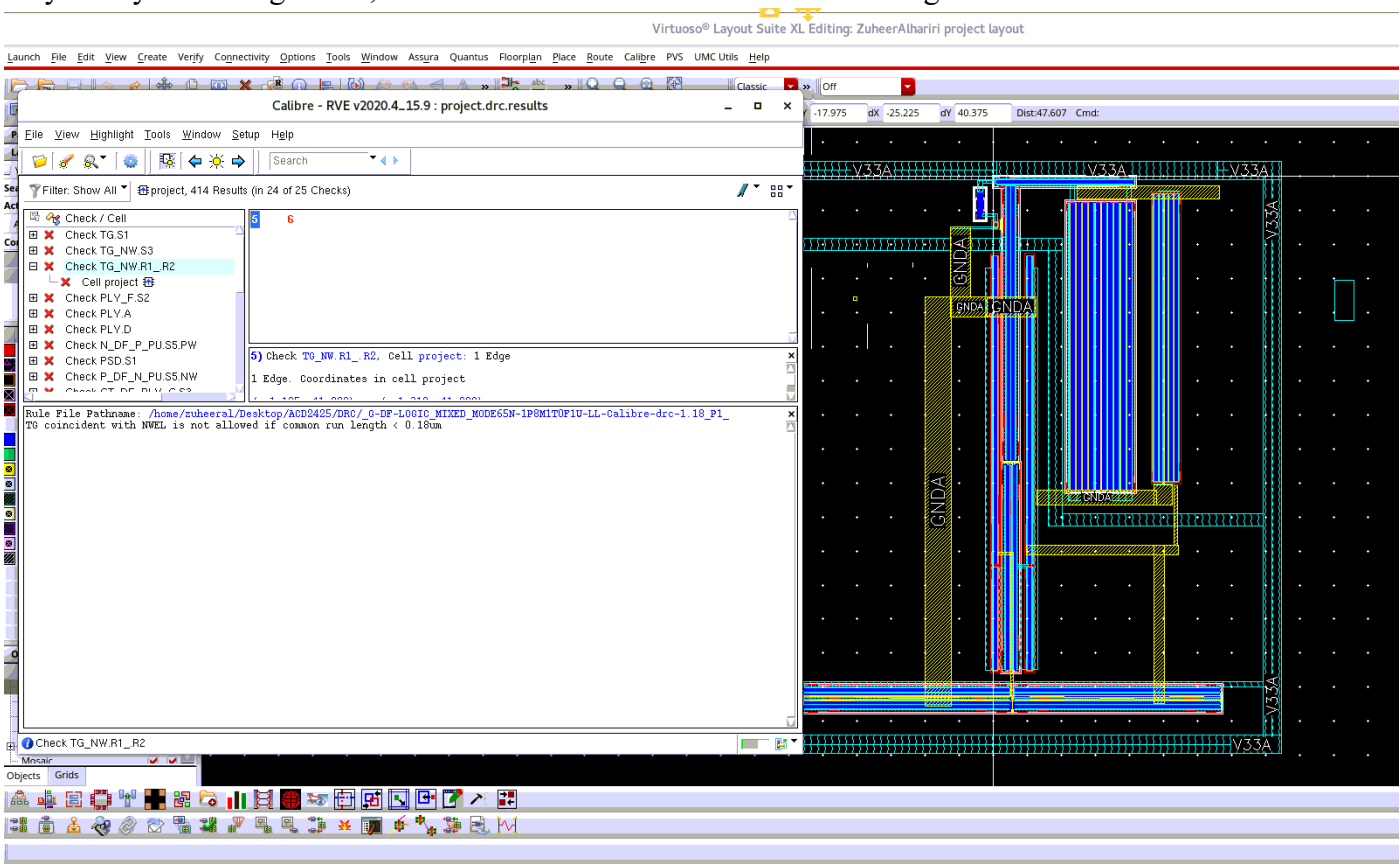

figure 6: running layout DRC

At this point LVS was not even attempted because this part needs to be revisited.

Post layout Simulation results:

Since the layout could not be completed the final results will be conducted with a 3.5pF load capacitance as shown below.

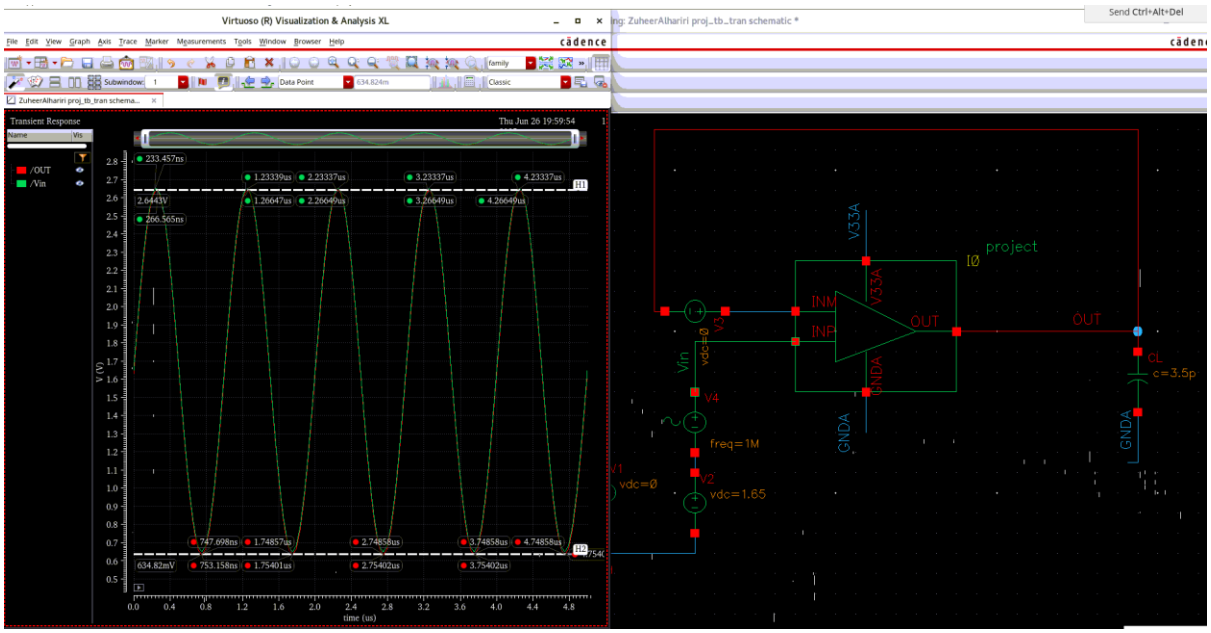

figure 7: transient analysis showing the swing of the output voltage

figure 8: Maximum over-shoot using 100ns period, 100mpp

The settling time can be calculated 6.6ns, which for a 100ns period is less than 10% meeting the criteria.

figure 9: slew rate with 100ns period, 1Vpp

Slew rate is calulated to be 1.37V/6.6ns or approximately 0.2V/ns